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HANOI UNIVERSITY OF INDUSTRY

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OPENING REMARKS

Fundamental And Applied IT Research (referred to as FAIR) is organized to contribute to promoting fundamental and applied research in Information Technology in Vietnam. This conference is under the chairmanship of the Vietnam Union of Science and Technology Associations, in collaboration with the Vietnam Academy of Science and Technology, Vietnam National University, Hanoi, and leading scientific institutions and researchers from research institutes and universities nationwide.

In 2025, the Conference Steering Committee, in partnership with the Hanoi University of Industry, organized the 18th National Scientific Conference (FAIR'2025) with the theme "Digital Transformation and Future Trends" on August 21–22, 2025, in Hanoi.

This conference serves as a forum for researchers in the field of Information Technology (IT), particularly faculty members from universities and research institutes across the country, to present their latest findings and exchange experiences in IT research and applications.

Currently, Vietnam is advancing its scientific and technological development, innovation, and digital transformation to become a high-income, developed nation. In this process, Information Technology, particularly Artificial Intelligence and communications plays a crucial role. We are building an economy based on high technology, with Vietnam actively undergoing digital transformation across all sectors. This process is unfolding dynamically and vigorously. With the theme "Digital Transformation and Future Trends", the conference acknowledges the substantial benefits of AI and digital transformation. We recognize that data has become an invaluable resource, yet the world is increasingly flooded with fake data and misinformation, largely generated by AI. As the IT community, we must actively engage in developing measures to mitigate these negative impacts, safeguarding users and ensuring societal safety.

The conference received over 170 submitted scientific reports addressing all current issues in Information Technology and Communications. The Program Committee conducted rigorous peer review and selection, accepting 112 papers for presentation in the conference's specialized sessions, 107 of which were selected for inclusion in the Conference Proceedings.

This year, the conference also hosted the FAIR PhD Forum 2025, attracting numerous doctoral candidates, graduate students, and young researchers. The Program Committee also selected the Best Paper Award for both research groups and young authors.

On behalf of the Organizing Committee and Program Committee, we extend our sincere gratitude to all authors who submitted their work to the conference and to the scientists who provided objective and valuable reviews of the submitted papers. We express our appreciation to the scientists who chaired the sessions in the specialized sessions and to all colleagues who participated in discussions. We are especially grateful to the Publishing House for Science and Technology, Vietnam Academy of Science and Technology, for their support in publishing the Conference Proceedings.

Finally, we extend our deepest gratitude to the Vietnam Union of Science and Technology Associations and the Hanoi University of Industry, the host institution for their significant efforts and time in organizing FAIR'2025. We also thank all sponsors whose multifaceted support and financial contributions were instrumental in ensuring the success of this conference.

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A Multi-Objective GA-LSTM Framework for Simultaneous Optimization of Predictive Accuracy and Computational Efficiency in Bitcoin Price Forecasting

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***Abstract*— Accurate forecasting of Bitcoin prices is challenging due to high volatility and nonlinear dynamics. This study proposes a GA-LSTM framework that integrates Long Short-Term Memory (LSTM) networks with Genetic Algorithm (GA)-based hyperparameter optimization to improve prediction accuracy and efficiency. The GA systematically tuned key hyperparameters, including layer count, neuron count, dropout rate, learning rate, batch size, and input window size. Experiments on historical Bitcoin data (Yahoo Finance, 2014–2025) showed that the GA-optimized LSTM outperformed the baseline, reducing MSE, RMSE, and MAE by 39.78%, 22.40%, and 24.37%, respectively. The approach also significantly reduced training time while maintaining high accuracy, highlighting the potential of evolutionary optimization in enhancing financial time series forecasting, especially in volatile markets.**

Keywords—Bitcoin price forecasting, Genetic Algorithm, LSTM, hyperparameters optimization.

I. INTRODUCTION

Cryptocurrencies, particularly Bitcoin [1], have garnered significant attention in recent years due to their decentralized nature, rapid growth, and high market volatility. Accurate forecasting of Bitcoin prices is of great importance to investors, traders, and policymakers, as it enables better risk management, informed decision-making, and the development of algorithmic trading strategies. However, the highly nonlinear and noisy nature of cryptocurrency markets poses substantial challenges for traditional forecasting techniques [2].

In the realm of time series forecasting, LSTM networks [3], a variant of recurrent neural networks (RNNs), have shown remarkable effectiveness in capturing long-range dependencies and sequential patterns in financial data. Despite their strong performance, the predictive accuracy of LSTM models is highly sensitive to the configuration of hyperparameters, such as the number of layers, number of neurons, dropout rate, learning rate, batch size, and input window size. Manually selecting these hyperparameters through trial-and-error or grid search methods is not only inefficient but often suboptimal, especially when dealing with high-dimensional search spaces.

To address this challenge, this study proposes using GA [4], a popular metaheuristic optimization technique inspired by natural selection, to automatically tune the hyperparameters of the LSTM model [5]. GA is particularly well-suited for navigating large, complex, and non-convex search spaces, making it an effective tool for deep learning model optimization.

The main contributions of this study include the development of a novel hybrid forecasting framework that integrates LSTM networks with GA-based hyperparameter optimization for Bitcoin price prediction. This framework leverages a multi-objective fitness function that simultaneously minimizes prediction error (Root Mean Squared Error – RMSE) and training time, enabling a more balanced and efficient model selection process. Additionally, a comprehensive experimental evaluation was conducted to compare the performance of the GA-optimized LSTM model (GA-LSTM) against a baseline LSTM model using manually selected hyperparameters.

The results demonstrate that the proposed GA-LSTM significantly outperforms the baseline model in terms of both accuracy and training efficiency. These findings highlight the potential of evolutionary algorithms in improving the performance of deep learning models for financial time series forecasting, particularly in the context of highly volatile and data-intensive markets like cryptocurrencies.

The remainder of this paper is organized as follows. Section 2 presents a review of related work on Bitcoin price prediction and hyperparameter optimization techniques in deep learning. Section 3 introduces the proposed GA-LSTM framework, detailing the integration of Genetic Algorithm for optimizing LSTM hyperparameters in the context of Bitcoin

* Corresponding Author.

price forecasting. Section 4 reports the experimental setup, evaluation metrics, and results, followed by a comprehensive discussion of the model's performance compared to the baseline. Finally, Section 5 concludes the paper and outlines potential directions for future research.

II. RELATED WORK

In recent years, deep learning models have garnered substantial interest in cryptocurrency price prediction due to their capacity to capture nonlinear patterns and temporal dependencies inherent in highly volatile, non-stationary markets. Among these models, LSTM networks have emerged as a dominant architecture, largely owing to their ability to retain long-term temporal features and address issues such as vanishing gradients in financial time series forecasting [6], [7].

A growing body of literature has investigated the application of LSTM and its variants in cryptocurrency markets. Wen and Ling [8], for instance, conducted a comparative study of LSTM and Convolutional Neural Networks (CNNs) on a five-year Bitcoin dataset, demonstrating that LSTM consistently outperformed CNNs with lower RMSE and higher R^2 scores across short-term forecasting windows. Similarly, Gu et al. [9] proposed a hybrid LSTM–AR(2) framework, which incorporated autoregressive components to enhance feature interpretation and reduce forecasting errors, thereby outperforming conventional LSTM models across various evaluation metrics. Kumar et al. [10] introduced a hybrid model combining a Beta seasonal autoregressive moving average model with LSTM, designed to capture both stochastic seasonal dynamics and nonlinear patterns. Their approach demonstrated superior predictive accuracy for both one-step and multi-step forecasts compared to ARIMA, LSTM, MLP, and ARIMA-ANN models across diverse datasets, including Bitcoin and other real and simulated time series.

In the context of input data processing, Koo and Kim [11] proposed a Centralized Clusters Distribution (CCD) filtering mechanism integrated with a Weighted Empirical Stretching (WES) loss function to improve Bitcoin price forecasting, particularly at the distribution tails. Their CCD-WES framework, when combined with LSTM and Singular Spectrum Analysis (SSA) decomposition, achieved notable improvements, with RMSE reductions of 11.5% overall and 22.5% in the extreme domain compared to baseline models. Gajamanage et al. [12] advanced this line of work by developing a dual LSTM framework for real-time financial forecasting. By iteratively training two LSTM networks—one to determine the optimal number of epochs and the other to generate forecasts—their method reduced error propagation and outperformed single LSTM, extended Kalman filter, and ARIMA models across stock, cryptocurrency, and commodity markets.

Further contributions have focused on improving the robustness of LSTM forecasts under uncertainty. Nguinabé et al. [13] introduced a bootstrap-based prediction interval framework for LSTM, enhancing point and interval forecast accuracy while reducing sensitivity to data assumptions. Their method, tested on hourly weather data, effectively complemented standard LSTM forecasts by capturing uncertainties across different forecasting horizons.

While LSTM-based models have demonstrated impressive predictive capabilities, their performance is highly dependent on appropriate hyperparameter tuning. This has spurred substantial research into automated hyperparameter optimization (HPO) techniques. Metaheuristic and evolutionary algorithms—including GA, Particle Swarm Optimization (PSO), and Bayesian Optimization (BO)—have been widely explored to address this challenge. Raiaan et al. [14] reviewed HPO strategies for CNNs and highlighted the limitations of manual tuning, while Kervancı et al. [15] demonstrated the superior performance of Bayesian-optimized LSTM, GRU, and hybrid architectures over grid and random search methods.

Hybrid approaches combining deep learning and optimization algorithms have also gained traction. Omole and Enke [16] employed a CNN–LSTM model with Boruta-based feature selection for Bitcoin price direction forecasting, achieving 82.44% accuracy and a simulated return exceeding 6600%. Singh et al. [17] integrated GA with various LSTM variants for stock market forecasting and reported an average accuracy improvement of over 60%, particularly for Bidirectional LSTM. Similarly, Sha [18] and Hafidi et al. [19] demonstrated the ability of GA-enhanced LSTM models to improve forecasting stability and accuracy in stock and cryptocurrency markets. More recently, Mehmed et al. [20] investigated GA and PSO for tuning ANN models in Bitcoin forecasting, with their GA-ANN hybrid achieving a validation accuracy of 97.54%, far surpassing the PSO-based alternative.

Despite these advances, prior studies have primarily focused on optimizing predictive accuracy while largely overlooking training efficiency—an essential factor for real-world deployment, particularly in high-frequency trading or resource-constrained environments. Existing GA-based HPO frameworks rarely consider model training time within their fitness functions, often resulting in highly accurate but computationally intensive models [17]– [20].

To address this gap, our study proposes a novel GA-LSTM framework that jointly optimizes both RMSE and training time through a multi-objective fitness function. By balancing predictive accuracy with computational efficiency, our approach offers a practical solution for real-time cryptocurrency forecasting, as demonstrated through comprehensive evaluations on Bitcoin price data. In contrast to earlier works, we provide rigorous baseline comparisons and show that our GA-optimized LSTM model not only enhances forecasting accuracy but also significantly reduces training time, thereby advancing the state of research on efficient deep learning for financial time series prediction.

III. GA-LSTM FOR BITCOIN PRICE FORECASTING

The workflow of the proposed method is illustrated in Fig. 1, which integrates a GA with a LSTM neural network for hyperparameter optimization in Bitcoin price forecasting. The process begins with the collection and preprocessing of historical Bitcoin data, which is then divided into training and testing sets. The GA initializes a population of candidate hyperparameter configurations, which are evaluated using a fitness function based on LSTM model performance. Through iterative genetic operations—selection, crossover, and mutation—the GA evolves towards more optimal solutions. This cycle repeats until a termination condition is met, at which point the best-performing hyperparameter set is selected. The optimized parameters are then applied to train the final LSTM model, which is evaluated on the testing set to determine its predictive performance and finalize the best configuration.

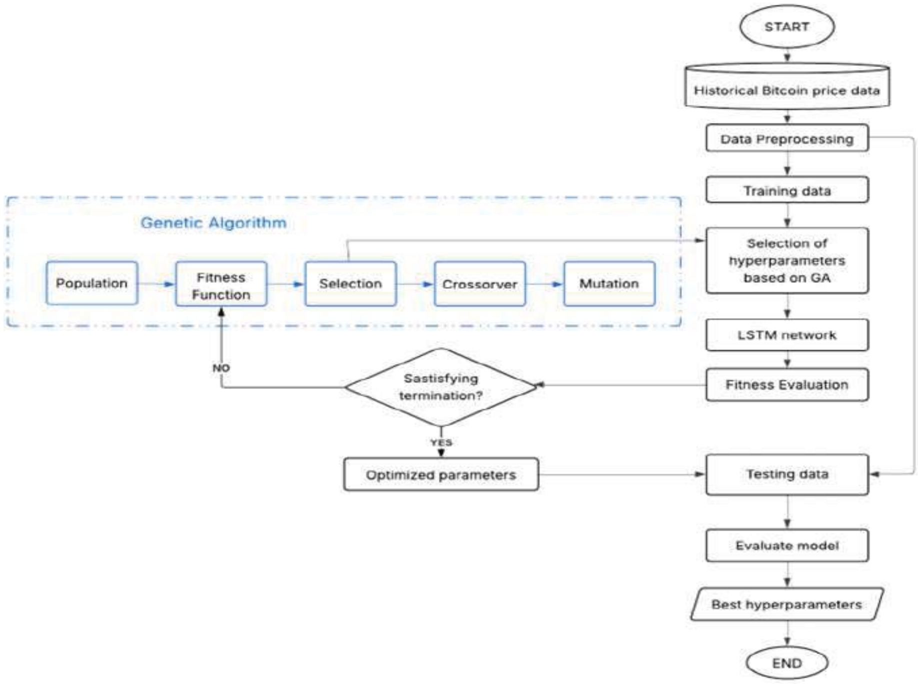

Fig. 1. The overall workflow of the proposed approach, where a Genetic Algorithm is used to optimize LSTM hyperparameters based on historical Bitcoin price data.

A. Data Collection and Preprocessing

Historical Bitcoin data was collected from Yahoo Finance using the yfinance Python library. The dataset includes features such as Open, High, Low, Close, and Volume, spanning from 17/9/2014 to 11/6/2025. The Close price was selected as the target variable for forecasting.

The preprocessing phase consisted of three key steps. First, all numerical features were normalized to the [0, 1] range using the MinMaxScaler to ensure uniform scaling across variables. Next, the sliding window technique was applied to convert the time series data into a supervised learning format, where input sequences of length window_size were constructed to predict values for the subsequent forecast_horizon days. Finally, the dataset was partitioned into training, validation, and testing sets following a 70:15:15 ratio to facilitate model training and evaluation.

B. LSTM Model Configuration

An LSTM network [21] was developed using the TensorFlow/Keras framework, with the baseline architecture comprising two stacked LSTM layers followed by a Dense output layer. The model's performance is highly dependent on several key hyperparameters, including the number of LSTM layers (n_lstm_layers), the number of neurons per layer (n_neurons), dropout rate (dropout_rate), learning rate (learning_rate), batch size (batch_size), input window size (window_size), and the number of training epochs (epochs). Careful tuning of these parameters is essential to achieve optimal forecasting accuracy.

C. Genetic algorithm for hyperparameter optimization

To efficiently explore the high-dimensional and complex search space of LSTM hyperparameters, we employed a GA, a population-based metaheuristic inspired by the principles of natural selection and evolutionary biology. The GA begins with an initial population of randomly generated individuals, where each individual encodes a unique set of hyperparameter values. These include the number of LSTM layers (n_lstm_layers), number of neurons per layer (n_neurons), dropout rate (dropout_rate), learning rate (learning_rate), batch size (batch_size), input window size (window_size), and number of training epochs (epochs).

The evolution of the population is guided by three genetic operators: selection, which chooses the fittest individuals based on the fitness function; crossover, which recombines parts of parent individuals to create offspring; and mutation, which introduces small random changes to promote genetic diversity. This iterative process continues over multiple generations, gradually improving the population's overall fitness.

In this study, the GA was configured with a population size of 50 (n_pop = 50) and evolved over 100 generations (n_gen = 100). To preserve high-quality solutions, the top 2 individuals were retained as elites (elite = 2) in each generation. The crossover probability was set to 0.8 (cxpb = 0.8), promoting diversity through recombination, while the mutation probability was set to 0.1 (mutpb = 0.1) to introduce controlled random variations and prevent premature convergence.

A multi-objective fitness function was designed to guide the optimization process by simultaneously minimizing both prediction error and computational cost. Specifically, the objectives are:

RMSEval(θ): Root Mean Squared Error on the validation set, measuring prediction accuracy.

TrainingTime(θ): Total time required to train the model with hyperparameter configuration θ.

The optimization goal is to identify a set of Pareto-optimal solutions as described in (1).

$$\min_{\theta \in \Theta} F(\theta) = [RMSEval(\theta), TrainingTime(\theta)] \quad (1)$$

where θ denotes a candidate hyperparameter vector in the search space Θ.

Alternatively, a scalarized version of the fitness function using a weighted sum can be applied as shown in (2)

$$\min_{\theta \in \Theta} F_{weighted}(\theta) = \alpha . RMSE_{val}(\theta) + \beta . TrainingTime(\theta) \quad (2)$$

where α, $\beta \in [0,1]$ representing the relative importance of accuracy versus efficiency.

Algorithm 1 presents the pseudocode of GA-based Hyperparameter Optimization

Algorithm 1. GA-LSTM Hyperparameter Optimization for LSTM.

Input:
- Dataset D
- Population size P
- Number of generations G
- Hyperparameter search space Θ

Output:
- Best hyperparameter configuration θ_best

```
Initialize population Pop of size P with random individuals θ ∈ Θ
for generation = 1 to G do
    for each individual θ in P do
        Train LSTM model with θ on training data
        Evaluate fitness(θ) = [RMSE_val(θ), TrainingTime(θ)]
    end for
    Select parent individuals from P based on fitness
    Apply crossover and mutation to generate new offspring
    Create new population P from best individuals and offspring
end for
Return θ_best from P with lowest RMSE_val and acceptable TrainingTime
```

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents a detailed evaluation of the proposed GA-based hyperparameter optimization framework for LSTM in Bitcoin price forecasting. We assess the convergence of the genetic algorithm, analyze the trade-offs between performance and training time, and compare the GA-optimized model with a baseline LSTM using default parameters.

A. *Convergence behavior of Genetic Algorithm*

To analyze the learning efficiency of the GA, we tracked the RMSE across 100 generations. The convergence process is illustrated in Fig. 2, which shows that RMSE dropped significantly during the first 10 generations and gradually flattened after generation 20, indicating rapid early convergence followed by stability.

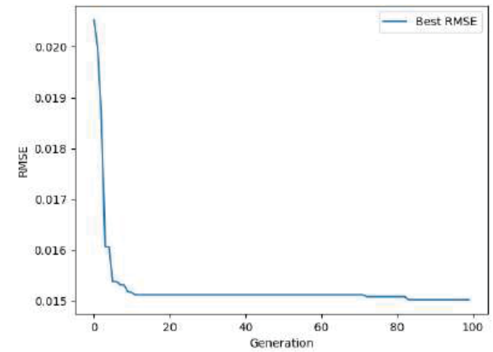

Fig. 2. Convergence of RMSE over 100 generations, showing rapid improvement during the first 10 generations followed by stabilization.

In parallel, Figure 3 shows the fluctuation in training time per generation. While earlier generations exhibit highly variable training durations (with peaks exceeding 30 seconds), the trend stabilizes as GA selects faster and more efficient solutions.

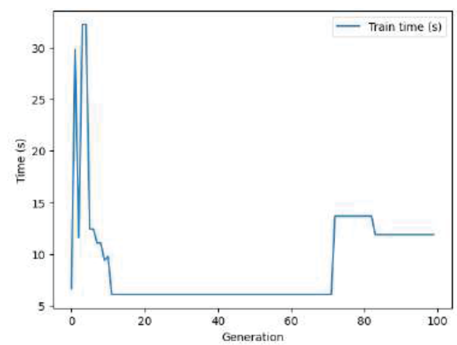

Fig. 3. Training time per generation during the GA process.

B. *Pareto front: Accuracy vs. Training time*

To evaluate the trade-off between model performance and computational efficiency, a Pareto front was constructed using the final population from the GA. As shown in Fig. 4, the curve highlights several non-dominated solutions that achieve favorable balances between low RMSE and reduced training time. These solutions allow researchers to select models based on available hardware or real-time constraints.

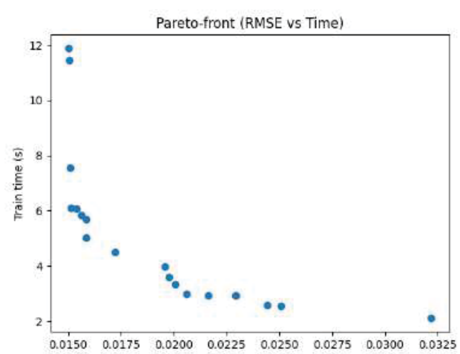

Fig. 4. Pareto front: RMSE vs. Training time.

C. Model performance comparison

The best-performing hyperparameter set identified by the GA is: n_lstm_layers = 2, n_neurons = 176, dropout_rate = 0.0025, learning_rate = 0.002843, batch_size = 37, epochs = 50. This configuration was selected as the final model for retraining on the full training dataset and evaluation on the test set.

The comparison between the GA-optimized model and the baseline LSTM (with fixed/default hyperparameters) is provided in Table 1.

Table 1. LSTM Performance Comparison

Metric	LSTM Vanilla	LSTM-GA Optimized	% Improvement
MSE	6761842	4071971	39.78%
RMSE	2600.354	2017.913	22.40%
MAE	1802.389	1363.12	24.37%
R2	0.989834	0.9938781	39.78%
Time (s)	53.74091	11.9	77.85% (Reduction)

These results clearly demonstrate that GA-based tuning significantly improves prediction accuracy while also reducing computational cost.

D. Visual comparison of Predicted vs. Actual values

Fig. 5 illustrates the predicted vs. actual closing prices of Bitcoin. The GA-optimized LSTM model more closely follows the actual price curve than the baseline model, especially in periods of rapid price fluctuation. This demonstrates its superior ability to learn temporal dependencies and adapt to volatile market behavior.

Fig. 5. Actual vs. Predicted Price – LSTM Vanilla vs. GA-Optimized.

E. Population analysis and Trade-off dynamics

The top-performing model achieved RMSE ≈ 0.015 with training time ~11.9 seconds, while other configurations traded minor losses in accuracy for up to 78% faster training. This supports the GA's role as a multi-objective optimizer rather than just a precision maximizer.

The experimental results demonstrate that the Genetic Algorithm (GA) rapidly converged to an optimal region of the hyperparameter search space within the first 20 generations. Compared to the baseline LSTM model, the GA-optimized variant achieved substantial improvements—approximately 40% reduction in MSE, 22% in RMSE, and 24% in MAE—highlighting its superior predictive accuracy. Moreover, the Pareto front revealed a diverse set of trade-off solutions balancing prediction accuracy and training time, offering flexibility for deployment in resource-constrained environments. Notably, the GA-enhanced LSTM exhibited better generalization on test data, particularly during periods of high market volatility.

V. CONCLUSION AND FUTURE WORK

This study proposed a hybrid GA-LSTM framework integrating Genetic Algorithm-based hyperparameter optimization with LSTM networks to improve Bitcoin price forecasting. The framework systematically tuned key hyperparameters via a multi-objective fitness function, achieving a balance between predictive accuracy and computational efficiency. The GA-optimized LSTM outperformed the baseline, reducing MSE, RMSE, and MAE by approximately 40%, 22%, and 24%, respectively. Pareto front analysis highlighted diverse trade-offs between accuracy and training time, while visual results demonstrated robustness under volatile market conditions. Future work may explore alternative metaheuristics (e.g., PSO, DE, Bayesian Optimization), advanced architectures (e.g., Bi-LSTM, GRU, Transformer), enriched features, and real-time deployment to further enhance performance and reliability in financial forecasting.

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REFERENCES

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System".

[2] A. Bouteska, M. Z. Abedin, P. Hajek, and K. Yuan, "Cryptocurrency price forecasting – A comparative analysis of ensemble learning and deep learning methods," International Review of Financial Analysis, vol. 92, p. 103055, Mar. 2024, doi: 10.1016/j.irfa.2023.103055.

[3] G. Van Houdt, C. Mosquera, and G. Nápoles, "A review on the long short-term memory model," Artif Intell Rev, vol. 53, no. 8, pp. 5929–5955, Dec. 2020, doi: 10.1007/s10462-020-09838-1.

[4] "Meta-Heuristic and Evolutionary Algorithms for Engineering Optimization 9781119387077," dokumen.pub. Accessed: Sep. 11, 2025. [Online]. Available: https://dokumen.pub/meta-heuristic-and-evolutionary-algorithms-for-engineering-optimization-9781119387077.html

[5] A. EL OUARDI, B. ER-RAHA, M. RIAD, and K. TATANE, "A gan-based method to tune lstm hyperparameters for financial forecasting," Journal of Theoretical and Applied Information Technology, vol. 103, no. 9, 2025, Accessed: Jun. 28, 2025. [Online]. Available: http://jatit.org/volumes/Vol103No9/17Vol103No9.pdf

[6] P. Bilokon and Y. Qiu, "Transformers versus LSTMs for electronic trading," Sep. 20, 2023, arXiv: arXiv:2309.11400. doi: 10.48550/arXiv.2309.11400.

[7] M. S. Sepehri, A. Mehradfar, M. Soltanolkotabi, and S. Avestimehr, "CryptoMamba: Leveraging State Space Models for Accurate Bitcoin Price Prediction," May 04, 2025, arXiv: arXiv:2501.01010. doi: 10.48550/arXiv.2501.01010.

[8] N. S. Wen and L. S. Ling, "Evaluation of Cryptocurrency Price Prediction Using LSTM and CNNs Models | Wen | JOIV : International Journal on Informatics Visualization," Nov. 2023, doi: http://dx.doi.org/10.30630/joiv.7.3-2.2344.

[9] C.-H. Wu, C.-C. Lu, Y.-F. Ma, and R.-S. Lu, "A New Forecasting Framework for Bitcoin Price with LSTM," in 2018 IEEE International Conference on Data Mining Workshops (ICDMW), Oct. 2018, pp. 168–175. doi: 10.1109/ICDMW.2018.00032.

[10] B. Kumar, Sunil, and N. Yadav, "A novel hybrid model combining βSARMA and LSTM for time series forecasting," Applied Soft Computing, vol. 134, p. 110019, Feb. 2023, doi: 10.1016/j.asoc.2023.110019.

[11] E. Koo and G. Kim, "Centralized decomposition approach in LSTM for Bitcoin price prediction," Expert Systems with Applications, vol. 237, p. 121401, Mar. 2024, doi: 10.1016/j.eswa.2023.121401.

[12] K. Gajamannage, Y. Park, and D. I. Jayathilake, “Real-time forecasting of time series in financial markets using sequentially trained dual-LSTMs,” Expert Systems with Applications, vol. 223, p. 119879, Aug. 2023, doi: 10.1016/j.eswa.2023.119879.

[13] J. Nguinabé, R. Rockefeller, N. M. Khan, and P. C. Rodrigues, “Bootstrap prediction intervals for the long short term memory (LSTM) recurrent neural network,” Expert Systems with Applications, vol. 284, p. 127728, Jul. 2025, doi: 10.1016/j.eswa.2025.127728.

[14] M. A. K. Raiaan et al., “A systematic review of hyperparameter optimization techniques in Convolutional Neural Networks,” Decision Analytics Journal, vol. 11, p. 100470, Jun. 2024, doi: 10.1016/j.dajour.2024.100470.

[15] I. S. Kervanci, M. F. Akay, and E. Özceylan, “Bitcoin price prediction using LSTM, GRU and hybrid LSTM-GRU with bayesian optimization, random search, and grid search for the next days,” JIMO, vol. 20, no. 2, pp. 570–588, Feb. 2024, doi: 10.3934/jimo.2023091.

[16] O. Omole and D. Enke, “Deep learning for Bitcoin price direction prediction: models and trading strategies empirically compared,” Financial Innovation, vol. 10, no. 1, p. 117, Aug. 2024, doi: 10.1186/s40854-024-00643-1.

[17] U. Singh, K. Saurabh, N. Trehan, R. Vyas, and O. Vyas, “GA-LSTM: Performance Optimization of LSTM driven Time Series Forecasting,” Computational Economics, pp. 1–36, Dec. 2024, doi: 10.1007/s10614-024-10769-0.

[18] X. Sha, “Time Series Stock Price Forecasting Based on Genetic Algorithm (GA)-Long Short-Term Memory Network (LSTM) Optimization,” May 06, 2024, arXiv: arXiv:2405.03151. doi: 10.48550/arXiv.2405.03151.

[19] N. Hafidi, Z. Khoudi, M. Nachaoui, and S. Lyaqini, “Cryptocurrency Price Prediction with Genetic Algorithm-based Hyperparameter Optimization,” Statistics Optimization & Information Computing, vol. 13, pp. 0–24, Feb. 2025, doi: 10.19139/soic-2310-5070-2035.

[20] A. B. Mehmet, “Hybrid Optimal Time Series Modeling for Cryptocurrency Price Prediction: Feature Selection, Structure and Hyperparameter Optimization.” Accessed: Jun. 25, 2025. [Online]. Available: https://search.trdizin.gov.tr/tr/yayin/detay/1267346

[21] “Long short-term memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: 10.1162/NECO.1997.9.8.1735.

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